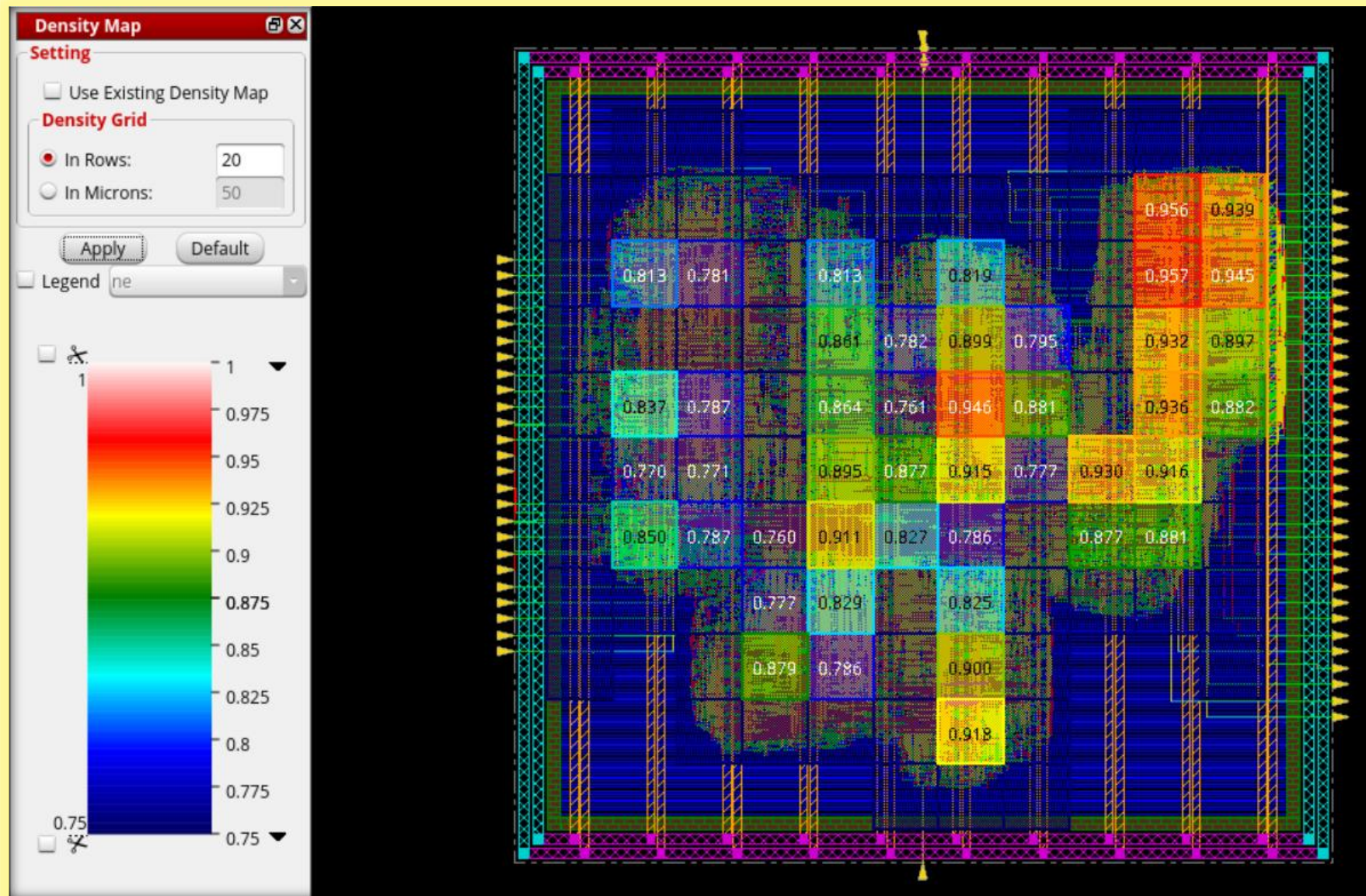


Fast Fourier Transform Processor (RTL to GDSII Implementation)



Timing report:

```
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Oct 18 2025  10:54:41 am
Module:           FFT
Operating conditions: PVT_0P9V_125C
Interconnect mode: global
Area mode:        physical library
=====

Path 1: MET (0 ps) Setup Check with Pin shift_2_shift_reg_i_reg[23]/CK->D
  Group: clk
  Startpoint: (R) rom8_s_count_reg[3]/CK
  Clock: (R) clk
  Endpoint: (R) shift_2_shift_reg_i_reg[23]/D
  Clock: (R) clk

      Capture      Launch
  Clock Edge: +    8000      0
  Src Latency: +      0      0
  Net Latency: +      0 (I)    0 (I)
  Arrival: =    8000      0

      Setup: -    146
  Uncertainty: -    10
  Required Time: = 7844
  Launch Clock: -      0
  Data Path: -    7844
  Slack: =      0
```

Power report:

```
Instance: /FFT
Power Unit: W
PDB Frames: /stim#0/frame#0

-----
Category      Leakage      Internal      Switching      Total      Row%
-----
memory        0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
register       4.34086e-07  1.56635e-03  5.21972e-04  2.08876e-03  14.21%
  latch       0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
  logic       3.21900e-06  4.47958e-03  8.12784e-03  1.26106e-02  85.79%
  bbox       0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
  clock       0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
  pad         0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
  pm          0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
-----
Subtotal      3.65309e-06  6.04593e-03  8.64981e-03  1.46994e-02  100.00%
Percentage    0.02%      41.13%      58.84%      100.00%  100.00%
-----
```

Area report:

```
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Oct 18 2025  10:54:44 am
Module:           FFT
Operating conditions: PVT_0P9V_125C
Interconnect mode: global
Area mode:        physical library
=====

Instance Module  Cell Count  Cell Area  Net Area  Total Area
-----
FFT              27423  107961.534  33607.027  141568.561
```

Quality of Result report:

Timing				

Clock Period				

clk	8000.0			

Cost Group	Critical Path Slack	TNS	Violating Paths	

clk	0.0	0.0	0	
default	No paths	0.0		

Total		0.0	0	

Instance Count				

Leaf Instance Count			27423	
Physical Instance count			0	
Sequential Instance Count			2358	
Combinational Instance Count			25065	
Hierarchical Instance Count			0	

Area				

Cell Area			107961.534	
Physical Cell Area			0.000	
Total Cell Area (Cell+Physical)			107961.534	
Net Area			33607.027	
Total Area (Cell+Physical+Net)			141568.561	

Max Fanout			2358 (clk)	
Min Fanout			0 (shift_16_valid)	
Average Fanout			2.3	
Terms to net ratio			3.3508	
Terms to instance ratio			3.5212	
Runtime			3963.0191130000003 seconds	
Elapsed Runtime			3979 seconds	
Genus peak memory usage			2658.51	
Innovus peak memory usage			no_value	
Postname			vlsicad	

SYNTHESIS

Gate count report: Summary

Type	Instances	Area	Area %

sequential	2358	19893.798	18.4
inverter	3839	6904.638	6.4
buffer	677	2291.058	2.1
logic	20549	78872.040	73.1
physical_cells	0	0.000	0.0

total	27423	107961.534	100.0

Lint report: Summary

Lint summary	
Unconnected/logic driven clocks	0
Sequential data pins driven by a clock signal	0
Sequential clock pins without clock waveform	4
Sequential clock pins with multiple clock waveforms	0
Generated clocks without clock waveform	0
Generated clocks with incompatible options	0
Generated clocks with multi-master clock	0
Paths constrained with different clocks	0
Loop-breaking cells for combinational feedback	0
Nets with multiple drivers	0
Timing exceptions with no effect	0
Suspicious multi_cycle exceptions	0
Pins/ports with conflicting case constants	0
Inputs without clocked external delays	26
Outputs without clocked external delays	33
Inputs without external driver/transition	26
Outputs without external load	33
Exceptions with invalid timing start-/endpoints	0
Total:	122

PnR: Pin placement

Pin placement summary:

checkPinAssignment Summary											
Partition	pads	pins	legal	illegal	internal	internal illegal	FT	FT illegal	constant	unplaced	
FFT	0	60	60	0	0		0	0	0	0	
TOTAL	0	60	60	0	0		0	0	0	0	

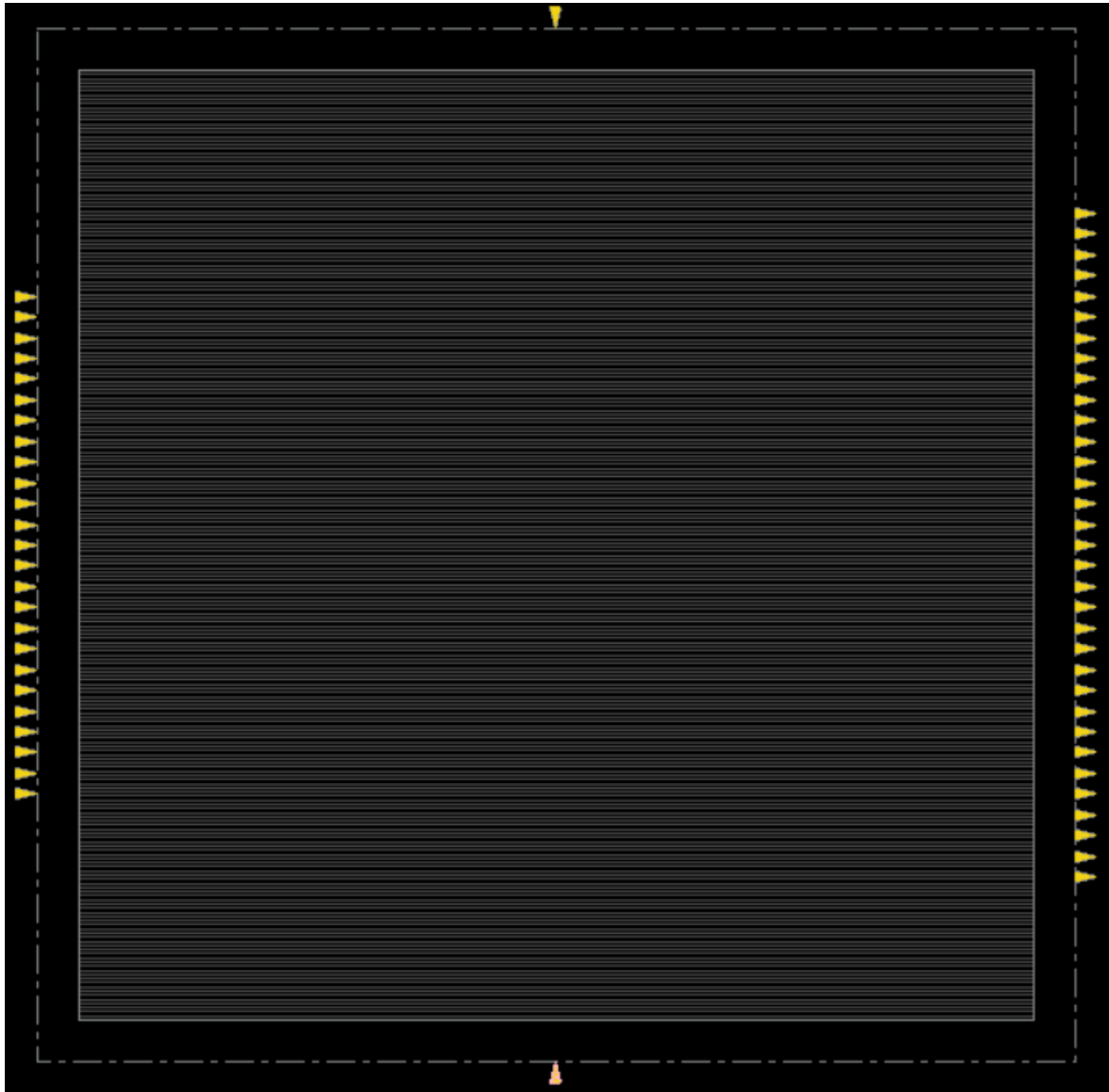
Pin placement details:

Direction	Signal Type	Metal Layer	Spacing	Description
Left	Inputs	M3	8.55 μ m	Inputs pins arranged vertically on left edge
Right	Outputs	M3	8.55 μ m	Outputs pins arranged vertically on right edge
Top	Clock (Clk)	M4	8.55 μ m	Clock pins placed on top edge
Bottom	Reset	M4	8.55 μ m	Reset pins placed on bottom edge

Note:

- 1. Inputs, Outputs and Reset are use as signal, and Clock use as clock signal
- 2. All pins are assign as fixed.

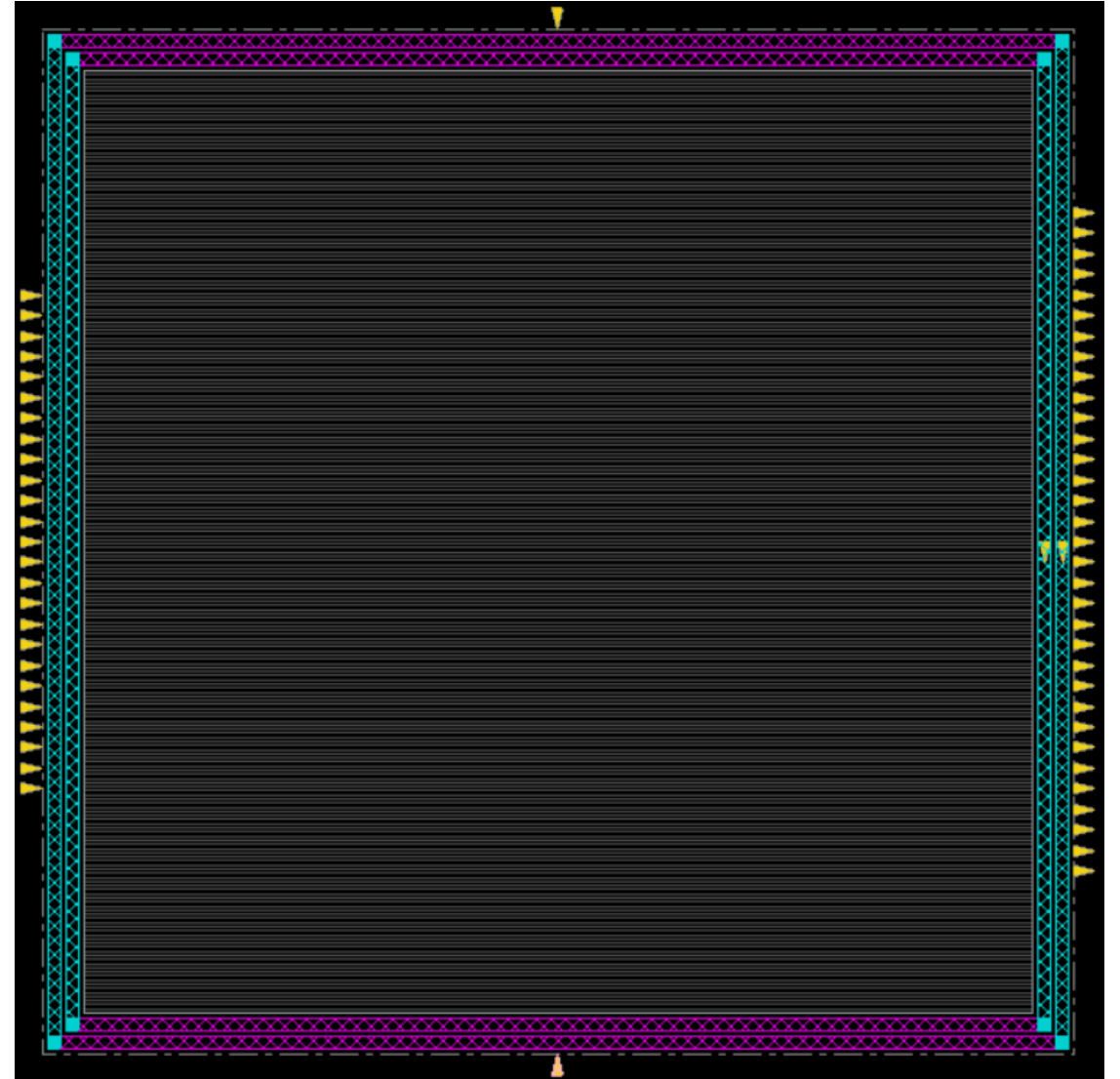
Pin placement:



PnR: Power Ring

A core power ring was created around the block using Add Rings in Innovus 22.10. Nets: VDD/VSS with Core ring(s) contouring around the core boundary.

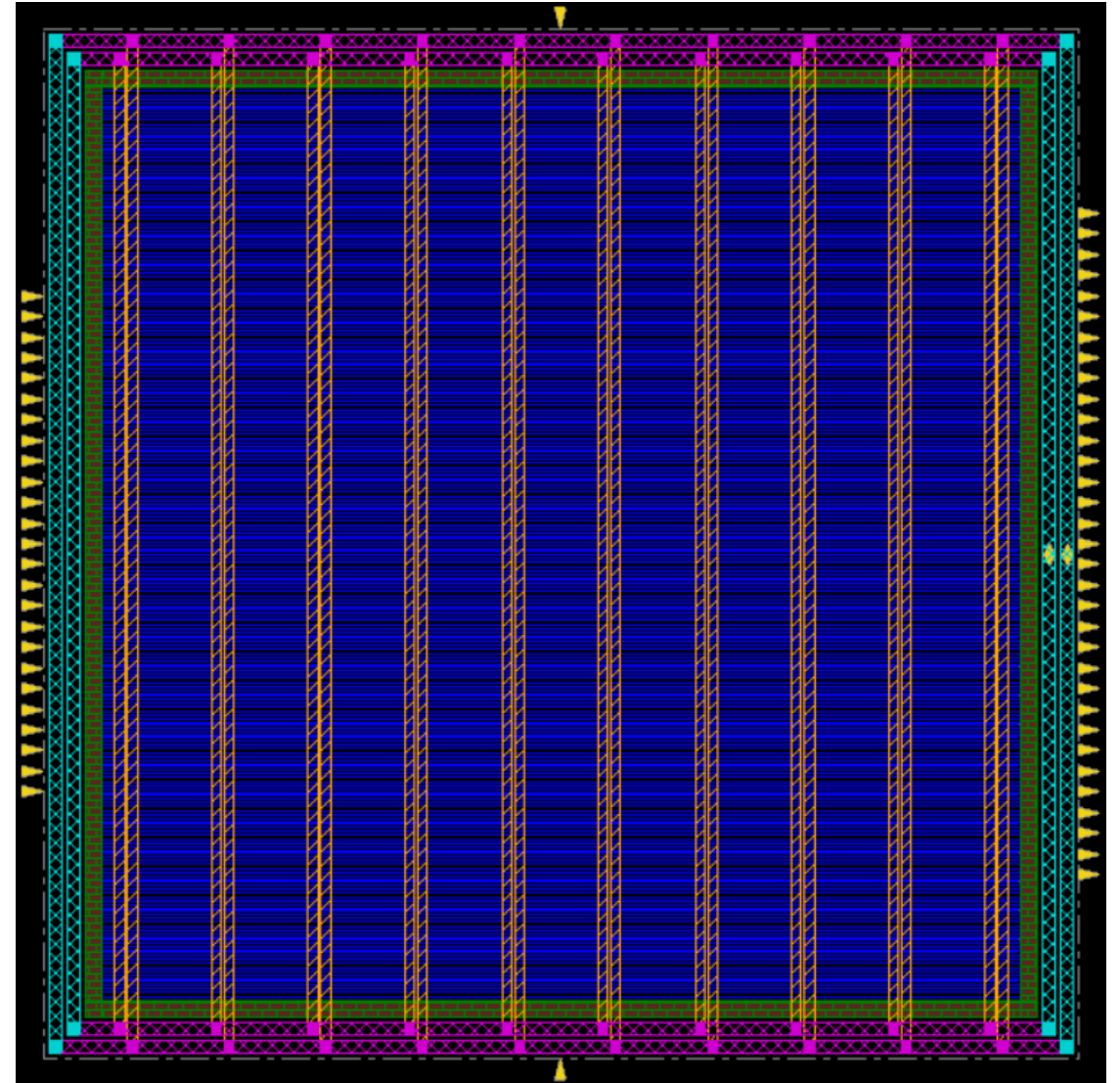
- **Top & Bottom: Metal7, width 5 μm**
- **Left & Right: Metal8, width 5 μm**
- **Ring spacing: 2.5 μm , offset: 1.8 μm (centered in the channel)**



PnR: Power Plan

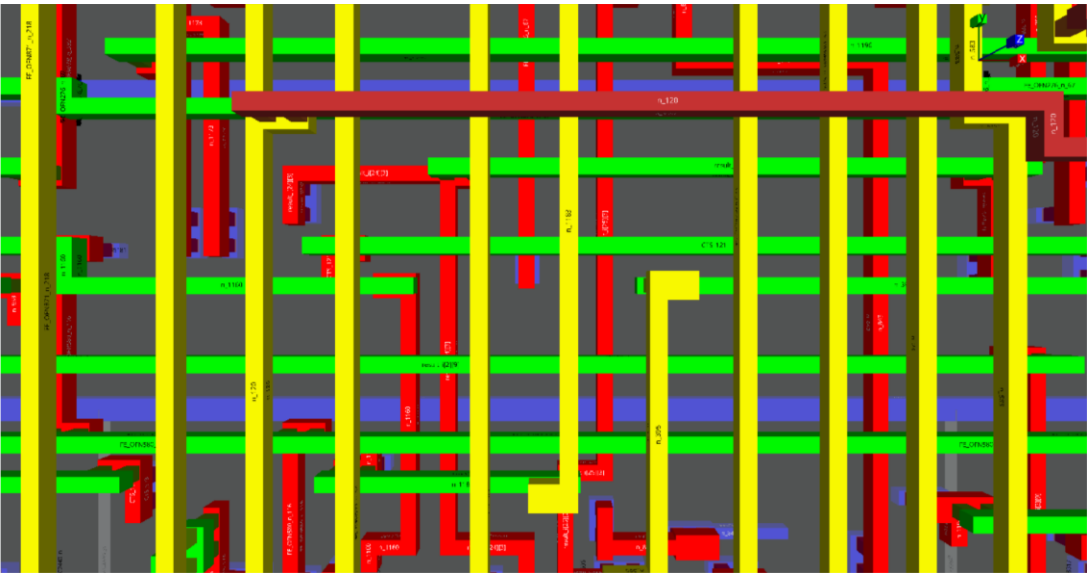
A global power grid was built after the ring:

- **Ring layers: M7/M8** around the core.
- **Core stripes: Vertical M6** VDD/VSS pairs across the core; **width 4 μm** each with **1.25 μm** intra-pair spacing, **offset 12.35 μm** from the left edge, and **40 μm pitch** across the block.
- **Stitching:** Stripes are stitched to the ring with via stacks (**top M7, bottom M6**).
- **Cell rail hookup: s-route on M1** ties standard-cell rails to the M6 stripes and the ring.

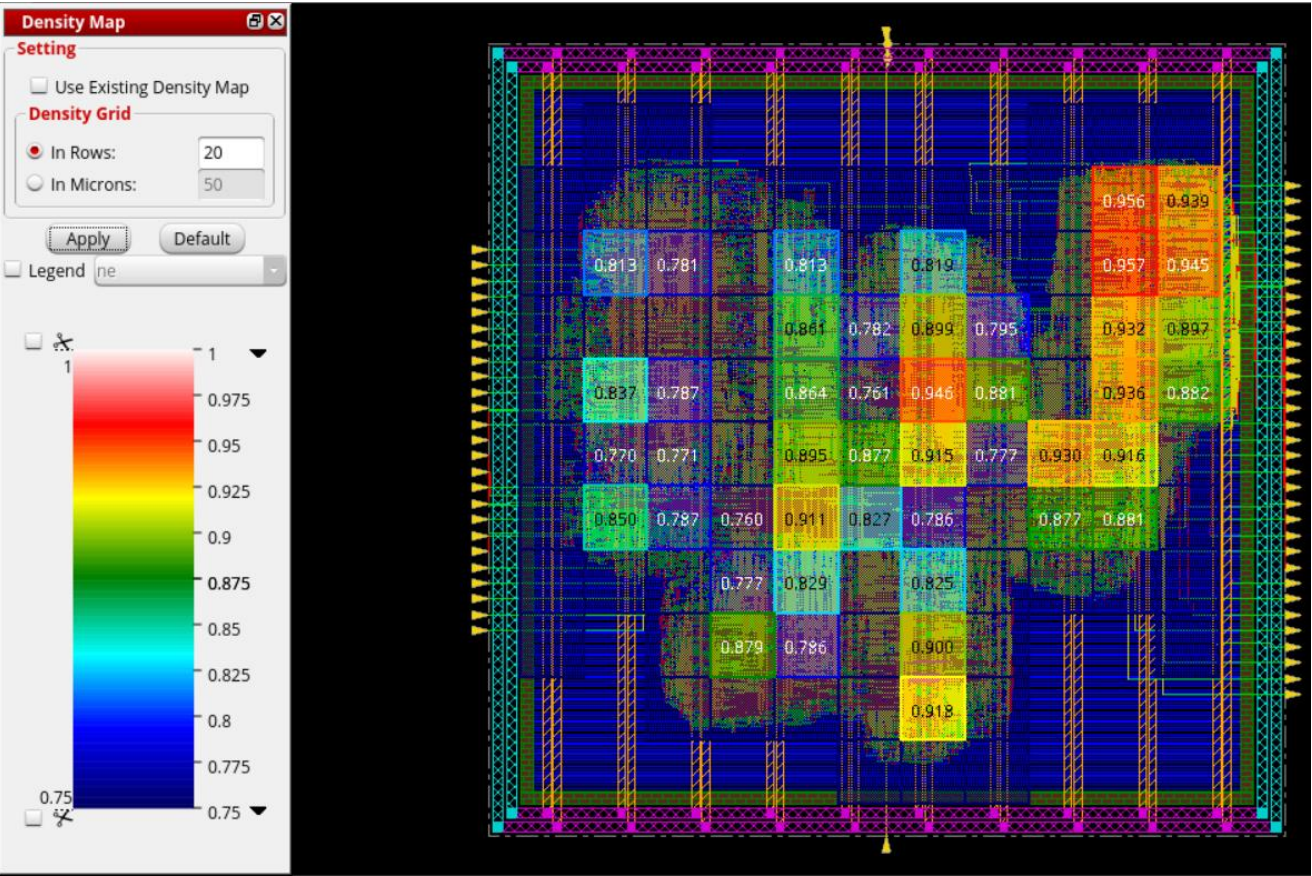


Routing Optimization (RO): Checklist

Layout: 3D view



Routing Density Map:



Routing Optimization (RO): Checklist

DRV checking report:

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)
max_length	0 (0)	0	0 (0)

Post-route setup check:

Setup mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	-0.003	-0.003	6.422	N/A	N/A	0.000
TNS (ns):	-0.003	-0.003	0.000	N/A	N/A	0.000
Violating Paths:	1	1	0	N/A	N/A	0
All Paths:	9164	6781	2383	N/A	N/A	0

Post-route hold check:

Hold mode	all	reg2reg	in2reg	reg2out	in2out	default
WNS (ns):	0.003	0.043	0.003	N/A	N/A	0.000
TNS (ns):	0.000	0.000	0.000	N/A	N/A	0.000
Violating Paths:	0	0	0	N/A	N/A	0
All Paths:	9164	6781	2383	N/A	N/A	0

Verify DRC :

Verification Complete : 0 Viols.

VERIFY CONNECTIVITY

```
Design Name: FFT
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (428.8000, 426.5500)
Error Limit = 100000; Warning Limit = 50
Check all nets
**** 07:15:16 **** Processed 5000 nets.
**** 07:15:16 **** Processed 10000 nets.
**** 07:15:17 **** Processed 15000 nets.
**** 07:15:17 **** Processed 20000 nets.
**** 07:15:17 **** Processed 25000 nets.
Net VDD: dangling Wire.
Net VSS: dangling Wire.

Begin Summary
  438 Problem(s) (IMPVFC-94): The net has dangling wire(s).
  438 total info(s) created.
End Summary

End Time: Tue Oct 28 07:15:18 2025
Time Elapsed: 0:00:02.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 438 Viols.  0 Wrngs.
```

VERIFY ANTENNA

```
***** START VERIFY ANTENNA *****
Report File: FFT.antenna.rpt
LEF Macro File: FFT.antenna.lef
5000 nets processed: 0 violations
10000 nets processed: 0 violations
15000 nets processed: 0 violations
20000 nets processed: 0 violations
25000 nets processed: 0 violations
Verification Complete: 0 Violations
***** DONE VERIFY ANTENNA *****
```